ID804 (RGB LED Driver IC for Smart Ambient Light)

Features

- AEC-Q100 Grade 2 Qualified
- AEC-Q102-003 Qualified
- Max 4KHz, 12-bit PWM Dimming Resolution
- 8-bit brightness resolution for R/G/B LED
- Maximum of 4078 devices in one LED chain
- Bidirectional, half-duplex, 2Mbps, CRC protected serial communication
- Support for 16 multicast address groups
- On-die oscillator
- Built-in diagnostic functions

General Description

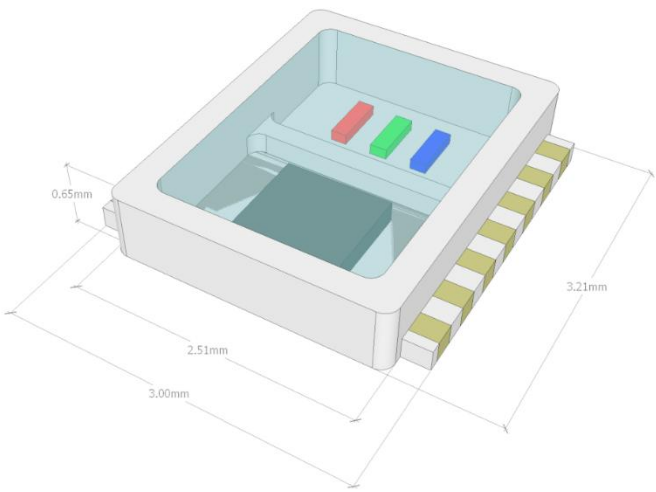
This device is designed to intelligently respond to the sophisticated ambient lighting scenarios in evolving automotive interiors. It incorporates a fully calibrated package containing three-color RGB LED chips and drive IC, ensuring precise calibration before shipment. Through a single microcontroller, up to 4078 devices can address and control each device in a daisy chain architecture via an open system protocol. Moreover, the microcontroller can detect the temperature and fault status of each device. The addition of CAN transceivers between daisy-chained devices ensures seamless long-distance data communication.

Applications

- Ambient Light
- Converting of single ended to differential signals

Ordering Information

Product ID	*Package*	*Body Size*
ID804	DFN3225-14L	3.21x3.0x0.65mm



Revision History

Date (YYYY-MM-DD)	Ver.	Editor	Contents
2025/06/16	0.70	K.LEE WS.KANG	Updated for: 1. Pin 1 Name change to (TM1) 2. Remove TC of Red Driving current 3. Electrical parameter table 4. Equation of Junction temp. Monitor 5. Fault diagnosis threshold information on description of register table 6. Electro-optical Characteristic table 7. Thermal Considerations
2025/06/25	0.71	K.LEE	Updated for: 1. Pin Map diagram change to TM1 2. Recommended Solder Pad Layout
2025/07/15	0.72	YS.KIM	Updated for: 1. Application Circuit Pin number typo
2025/07/15	0.73	K.LEE	Updated for: 1. Rotate the symbol in 2.1 Pin Map Top view 2. Added a decoupling capacitor in Application circuit
2025/10/20	0.74	K.LEE	Updated for: 1. Ch.3.3 Electro-optical characteristic 2. LED Driving current characteristic
2025/11/07	0.80	K.LEE	Updated for: 1. Design information regarding sub-component electrical characteristics has been migrated to the application note
2025/11/26	0.81	K.LEE	Updated for: 1. Real thermal resistance junction to solder in 3.1 Absolute Maximum Ratings
2025/12/01	0.82	K.LEE	Updated for: 1. Ch.3.3 Electro-optical characteristic (Adjustment of blue channel current)
2026/2/23	0.83	K.LEE	Updated for:

Date (YYYY-MM-DD)	Ver.	Editor	Contents
			1. Ch.3.3 Electro-optical characteristic (Adjustment of PWM_MAX RGB current) 2. Ch.5.1 Package Outline Dimension 3. Ch.4.3.3 Temperature Monitor equation
2026/4/3	0.84	K.LEE	Updated for: 1. Ch.4.4.1 Collected of TYPO (V_{LED} → $V_{DRV\_R/G/B}$) 2. Ch.3.2 Operating ambient temp. 3. Ch.3.2 Sleep/Active Max. Current value 4. Ch.3.3 Electro-optical Char.
2026/5/13	0.85	K.LEE	Updated for: 1. Ch.4.1.2 POR Max. value (3.9V) 2. Ch.3.3 Electro-optical char.(LED current info.) has been included in Ch.3.2 Recommended Operating conditions.
2026/8/6	0.90	K.LEE	Updated for: 1. Ch.3.2 Active mode Max. value 2. Ch.3.4 LED Forward voltage @ 20mA

Contents

1. Block Diagram

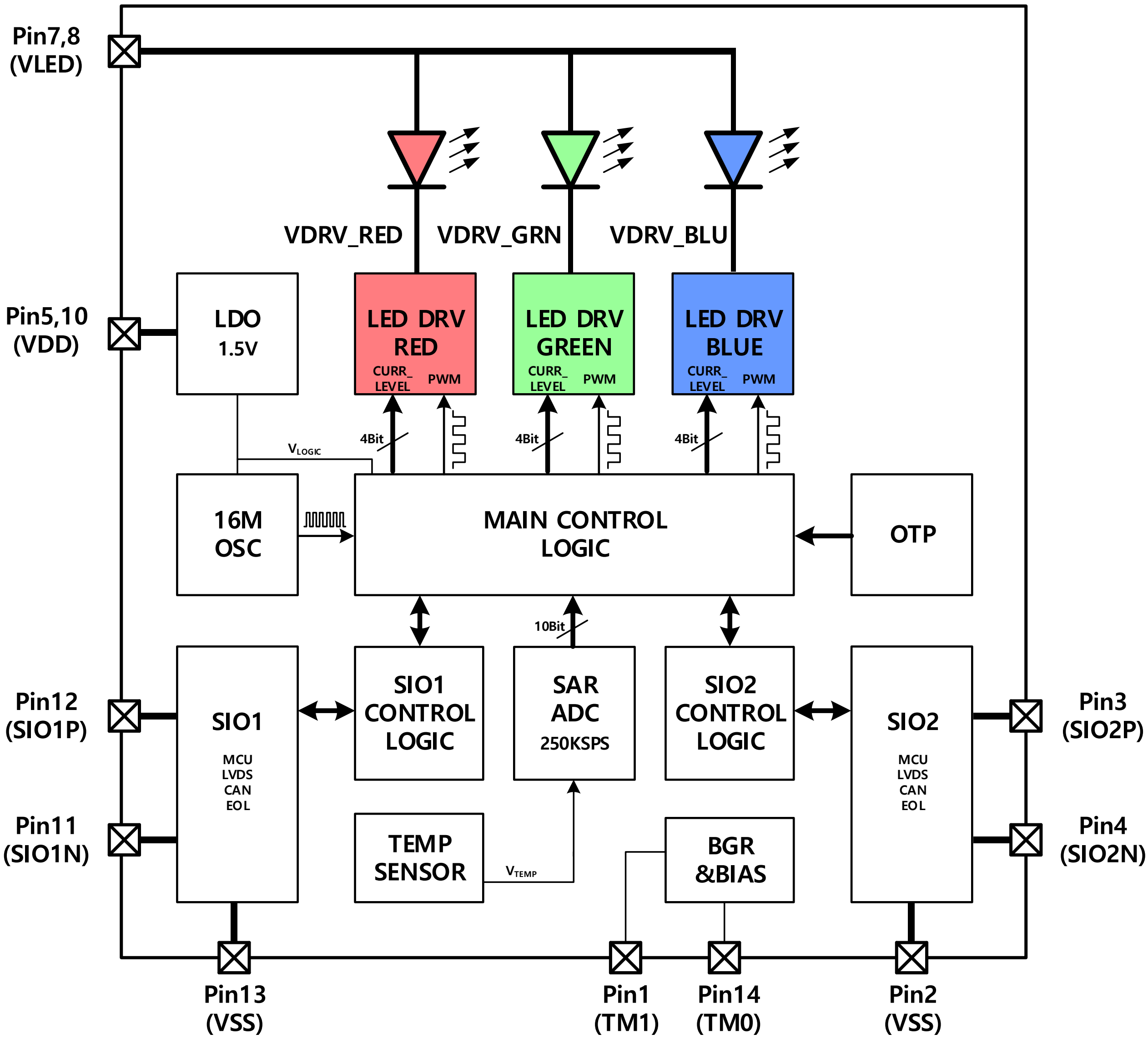


Figure 1. Functional Block diagram

The device implements a communication for the reception of control commands and for providing device status and configuration data. Low side, configurable constant current sinks are provided for controlling 3 LEDs (i.g. RGB). The main control logic computes the PWM duty cycles from the incoming commands and applies the corresponding control values to the three PWM generators. The main control logic is also in charge of a periodic temperature monitoring and an appropriate duty cycle adjustment for the red PWM channel. The actual device temperature is obtained via an integrated analog-digital converter (ADC). The die temperature is updated in the corresponding registers, and the values can be read by the MCU by host commands. As each device is individually calibrated to compensate for LED variations, the corresponding parameters can be stored in an one-time-programmable memory (OTP). The calibration data stored in the OTP is loaded into the main control logic whenever the IC is turned on or reset command.

2. Pin Configuration

2.1. Pin Map

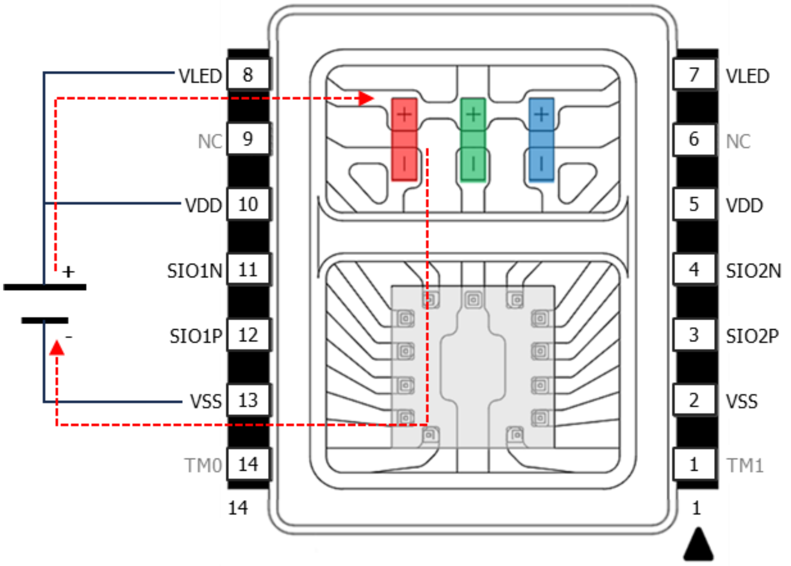


Figure 2. Top view with integrated LED and Driver IC

2.2. Pin Description

Pin No	Pin Name	Type	Description
1	TM1	IN	Connect to GND. (Test mode AMUX pin with a built-in pull-down resistor, 200KΩ)
2	VSS	GND	Ground
3	SIO2_P	IN/OUT	Serial communication interface 2 side, positive polarity
4	SIO2_N	IN/OUT	Serial communication interface 2 side, negative polarity
5	VDD	PWR	5V DC Supply voltage for IC
6	N.C	-	No connection (BLUE LED Cathode side)
7	VLED	PWR	5V DC Supply voltage for RGB LEDs
8	VLED	PWR	5V DC Supply voltage for RGB LEDs
9	N.C	-	No connection (RED LED Cathode side)
10	VDD	PWR	5V DC Supply voltage for IC
11	SIO1_N	IN/OUT	Serial communication interface 1 side, negative polarity
12	SIO1_P	IN/OUT	Serial communication interface 1 side, positive polarity
13	VSS	GND	Ground
14	TM0	I/O	Connect to GND (Test mode entry pin with a built-in pull-down resistor, 200KΩ)

3. General Product Characteristics

3.1. Absolute Maximum Rating

The absolute maximum ratings define values beyond which damage to the device may occur. Exposure to absolute maximum rating conditions for extended periods may affect device reliability. The functional operation of the device at these or any other conditions beyond the recommended operating ratings is not guaranteed.

Over operating free-air temperature range (unless otherwise noted)(1)

Parameter	Pin	MIN	MAX	UNIT
Input voltage range (with respect to GND unless specified)	VDD	-0.3	5.5	V
	VLED	-0.3	VDD+0.3	V
	TM0/1	-0.3	VDD+0.3	V
Input/Output voltage range (with respect to GND unless specified)	SIO1_P/N	-0.3	VDD+0.3	V
	SIO2_P/N	-0.3	VDD+0.3	V
Real thermal resistance junction to solder	Rth JS real (Typ. 93K/W)(2)		105	K/W
Soldering temperature	Reflow	260°C, 30 sec Max.		
Storage temperature of IC	T_{stg}	-40	125	°C
Junction temperature of IC	T_J	-40	125	°C
ESD	HBM(3)	±2000		V
	CDM(4)	±500		V
	CDM (Corner pins)	±750		V

(1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, and do not imply functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Conditions. Exposure to absolute- maximum-rated conditions for extended periods may affect device reliability.
(2) Color Set Point (255, 255, 255), Based on the drive current for a D65 white LED.
(3) AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001 specification.
(4) AEC Q100-011

3.2. Recommended Operating Condition

Parameter	Description	Min.	Typ	Max.	Unit
V_{VDD}, V_{LED}	DC supply voltage IC and LED	4.3	5	5.5	V
T_{RAMP_VDD}	VDD ramp-up time	0.5			ms
T_{INIT}	Device ready for commands, Initialization time after VDD > 4.3V			5.0	ms
f_{Comm_Data}	Allowed data rate for communication input	1.9	2.0	2.1	Mbps
$t_{msg\_interval}$	Time between two messages (send from MCU)		6		us
T_A	Operating ambient temperature	-40		110	℃
VDD Supply Currents (Test conditions are for T_{amb} = 25°C, VDD=5V, C_{VDD}=1μF)					
$I_{Q\_deepslp}$	Deep Sleep mode		600		uA
I_{Q_sleep}	Sleep mode, LED driver off, only IC		950	1100	uA
I_{Q_active}	Active mode, LED driver on, only IC*		1.1	1.3	mA
$I_{DRV\_D65\_Whtie}$	Active mode, LED driver on, only LED CURR_MAX_LVL(Addr=0x0E) = 0xBA3, w/(255,255,255)		33	39	mA

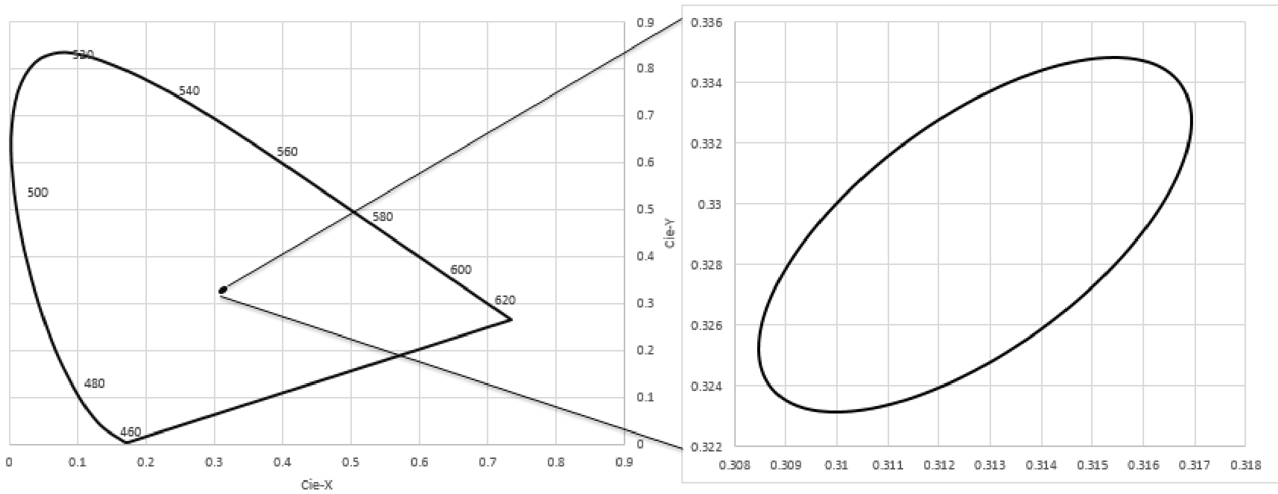
*Total ID804 SiP current consumption shall include LED drive current. “(Typ 1.1mA+i(LEDs)/200)+i(LEDs)”

3.3. Electro-optical Characteristic

Optical specifications are for T_{amb} = 25°C, VDD=5V, C_{VDD}=1μF, unless otherwise noted.

Parameter	Symbol	Color	Code	Conditions	Min.	Typ.	Max.	Unit
Dominant Wavelength (1)	λ_{D_R}	Red	W36	(255. 0, 0)	626	628	632	nm
	λ_{D_G}	Green	W25	(0, 255, 0)	528	530	534	nm
	λ_{D_B}	Blue	W12	(0, 0, 255)	458	462	464	nm
Luminous Intensity (2)	I_{v_R}	Red		(255. 0, 0)		690		mcd
	I_{v_G}	Green		(0, 255, 0)		1980		mcd
	I_{v_B}	Blue		(0, 0, 255)		130		mcd
	I_{v_W}	White (4)		(255,255,255)	-10%	2800	+10%	mcd
LED Forward Voltage	V_{F_R}	Red	D	@20mA		2.2		V
	V_{F_G}	Green	K	@20mA		2.6		V
	V_{F_B}	Blue	L	@20mA		2.8		V
View angle (3)	φ	-		(255,255,255)		120		degree

Note. (1) Dominant wavelength has an accuracy of ±2nm
(2) Luminous Intensity is tested by a tester calibrated by CAS-140B(CIE LED_B) and has an accuracy of 10%
(3) Viewing angle is the angle until 50% of brightness measured from the front part of LED.
(4) D65(6500K), The above luminous intensity represents brightness at 100%

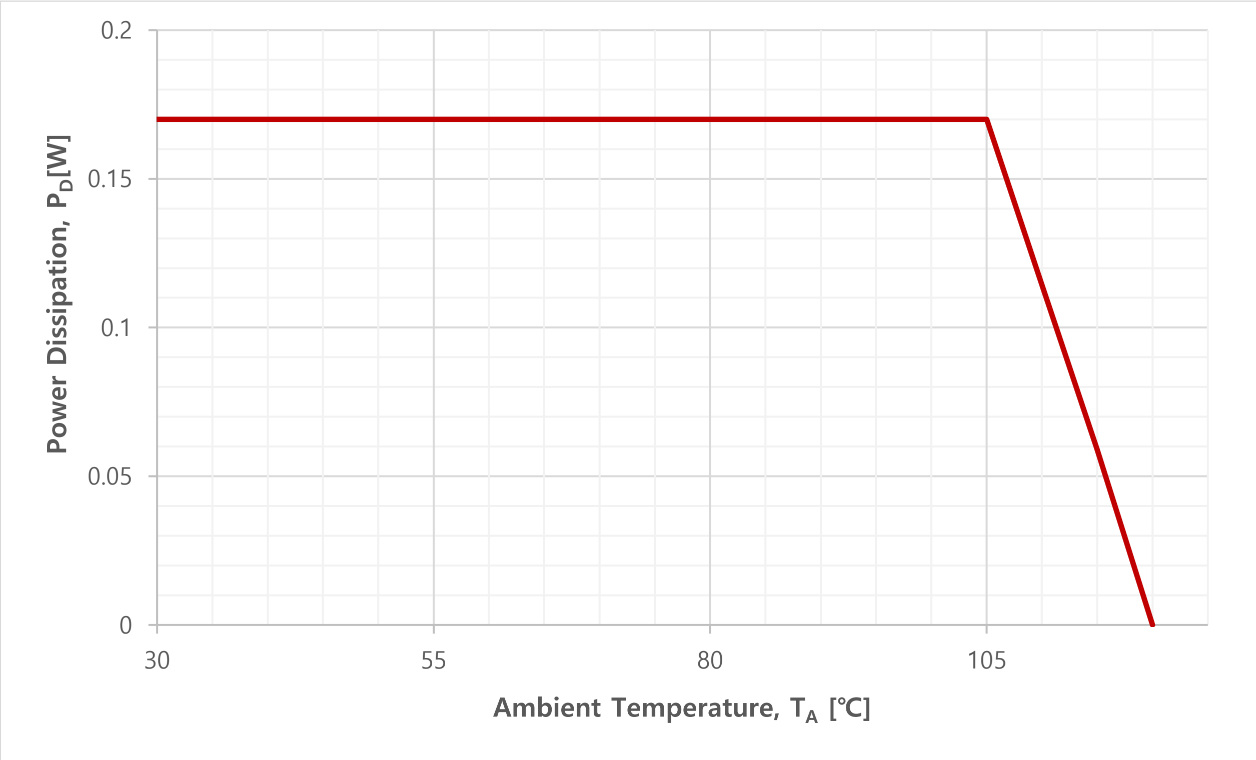


Rank	Description	CIE-x	CIE-y	a	b	Angle
D65	Macadams 3steps	0.3125	0.3290	0.00669	0.00285	58.57

3.4. Thermal Considerations

Need to suggest the maximum amount of power that the IC can consume under AEC-Q100 grade 2 conditions(T_A 105°C at still air). Thermal resistance characteristics would be variable depending on the IC, package, and PCB layer characteristics.

"ID804 of allowable IC power dissipation characteristics by ambient temperature"



4. Functional Description

4.1. Feature Description

4.1.1. Device States

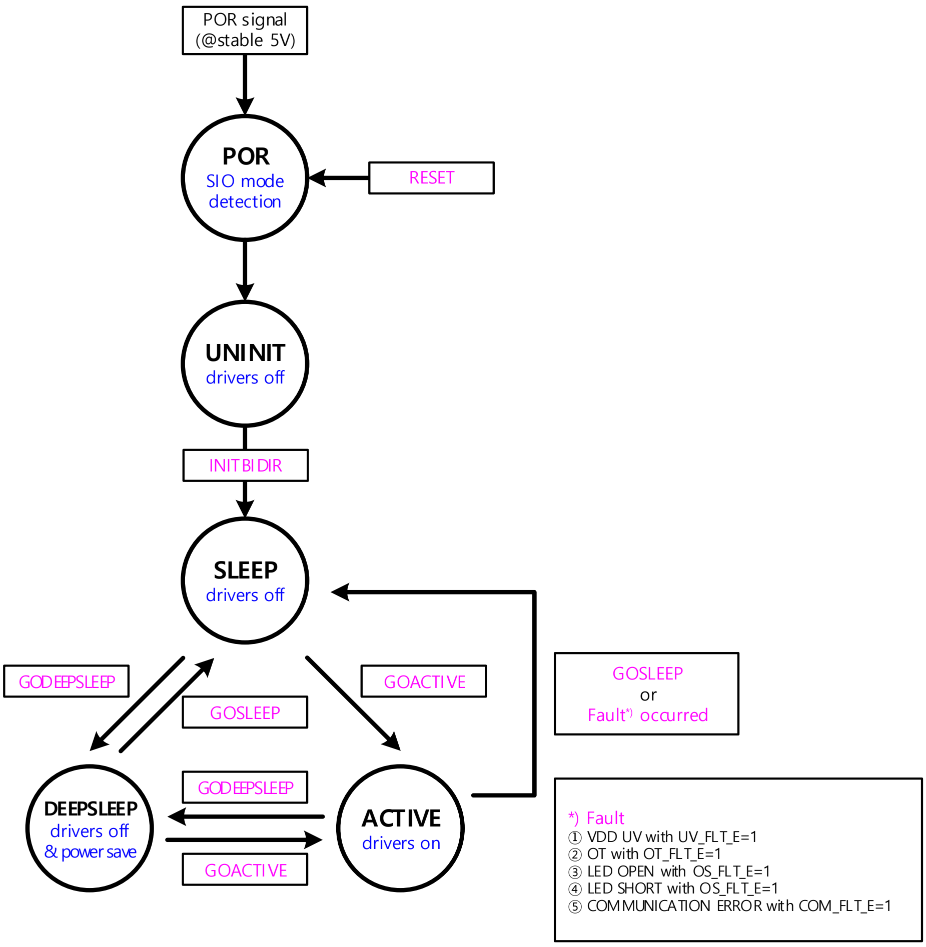


Figure 3. IC State

POR
: The ID804 is in an off state.

UNINIT
: The ID804 will reach this state automatically after power-up. PWM values are 0 and LED drivers disabled. In this state, the ID804 only responds to a limited set of commands:

- INITBIDIR
- RESET (broadcast)
- CLRERROR (broadcast)
- SET_SETUP1 (broadcast)
- SET_SETUP2 (broadcast)
- SET_TEMPTH (broadcast)
- SET_TEMPHYS (broadcast)
- SET_RGB (broadcast)

Other telegrams are ignored and fault flag (STATUS2.COM_FLT) set. Broadcast telegrams are fast forwarded and addressed telegrams are forwarded after the telegram has been fully received.

SLEEP

: The ID804 enters SLEEP state after receiving an INITBIDIR command, after a fault event (if selected) or via GOSLEEP command.
In this state, the LED drivers are disabled but the last PWM parameters are stored, i.e., not reset to 0. An update of the PWM parameters is possible.

ACTIVE

: The ID804 enters this mode after receiving the GOACTIVE command if no fault conditions are present.
In this state, the LED drivers are enabled with the respective current setting, PWM frequency and duty cycle for each channel. An update of the PWM parameters is possible.
After entering the ACTIVE state and LED drivers enabled, the open/short detection is executed for every channel with a PWM value exceeding the minimum value. See Chapter 4.4.

DEEPSLEEP

: The ID804 is in power save mode with reduced current consumption.
In this state, the ADC is disabled, and the LVDS receiver waits in a low power mode. When a signal is detected on the LVDS, the receiver switches to normal mode, transmits the data to the internal logic, and then returns to low power mode until the next LVDS signal is received.

Note. It is not recommended any other command send by MCU except state transfer command. (GOSLEEP, GOACTIVE)

4.1.2. Device Enable

When the supply voltage is connected and VDD increases above the V_{POR} level, or the RESET command is received, the ID804 undergoes the following startup sequence (Figure 4. Startup behavior of the ID804. The example shows the MCU (left) and EOL (right) modes. After a RESET command, the device starts in the PoR state.)

1. All registers are initialized to their default value (PoR in Figure 4).
2. Trim settings are loaded and checked for integrity. It takes max 120us. If a mismatch is detected, the OTPCRC flag in the STATUS2 register is set.
3. The voltage levels at each SIOx pin are checked to detect the communication modes (shaded signals in Figure 4).
4. ID804 enters the UNINIT state and is now ready to receive the first command (UNINIT in Figure 4).

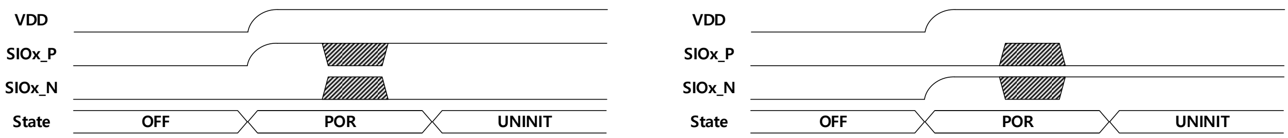


Figure 4. Startup behavior of the ID804. The example shows the MCU (left) and EOL (right) modes. After a RESET command, the device starts in the PoR state.

The whole sequence takes max. 5 ms.
(When sequence is started caused by RESET command, the whole sequence takes approximately 1 ms.)

Parameter	Description	Min.	Typ.	Max.	Unit
V_{POR}	Power-On-Reset (POR)			3.9	V

4.2. Communication

4.2.1. Single Ended, LVDS Link and Data Structure

4.2.1.1. Overview

- Bi-directional
- LVDS - differential Manchester coded.
- Single ended – Manchester coded & Manchester decoded or NRZ(clock & data), Push-pull interface, clock is invertible.
- Communication direction and SIOx connection is flexible.
- Communication speed is typ. 2Mbps Manchester coded.

4.2.1.2. Communication modes

Each one of the 2 available SIOx interfaces do support 4 different modes.
Each SIOx only can operate in semi-duplex mode meaning it either does receive or transmit a telegram at the same time. However, it is possible that a device does receive a telegram at one SIOx and start transmitting at the other SIOx interface.

The 4 communication modes are:

1. "LVDS-mode":
   The SIOx communication is using LVDS signaling for transmitting and receiving a communication telegram. The bit stream is using Manchester encoding to transfer DATA & CLK on the same line.
2. "MCU-mode":
   In this mode a SIOx is using single-ended mode for the signaling. Received telegrams are Manchester encoded on SIOx_P port. For transmitting, the device is decoding the Manchester signal. SIOx_P will deliver the DATA in NRZ format and SIOx_N delivers the corresponding clock.
3. "EOL(End Of Line)-mode":
   SIOx port does consider itself to be at the end of the daisy chain.
4. "CAN-mode":
   In this mode, it is possible to communicate via a CAN FD transceiver. SIOx_P is always in output state and transmits telegram as Manchester encoded signals (single-ended), SIOx_N is always in input state and receives telegrams as Manchester encoded signal (single-ended).

4.2.2. Communication mode selection

Communication mode auto detection takes place during initialization phase via voltage level check on SIOx_P & SIOx_N and this mode cannot be changed during the operation cycle. After a HW Reset (POR) or RESET command received the Initialization with mode selection take place again.

Table 1. Communication mode selection

Mode	Pin Configuration		I/O		Signal Type	Data Encoding	
	SIOx_P	SIOx_N	SIOx_P	SIOx_N		TX	RX
MCU	Pull-Up	Pull-Down	Bi-direction	Output or Hi-Z	Single-ended	NRZ (CLK&Data)	Manchester
LVDS	Pull-Down (By Internal Resistor)	Pull-Down (By Internal Resistor)	Bi-direction	Bi-direction	LVDS	Manchester	Manchester
CAN	Pull-Up	Pull-Up	Output	Input	Single-ended	Manchester	Manchester
EOL	Pull-Down	Pull-Up	-	-	-	-	-

4.2.3. MCU-mode communication physical layer and bit coding

Single ended communication in the master to slave direction is performed with so-called Manchester encoded. In that case a bit stream is sent via SIOx_P only; SIOx_N of the device is ignored. Slave to master communication is performed with NRZ coding. In that case SIOx_N provides a clock signal. The clock signal is inverted if it's selected via setup register. The typical communication data rate is 2Mbps.

4.2.3.1. Example of Communication Waveforms

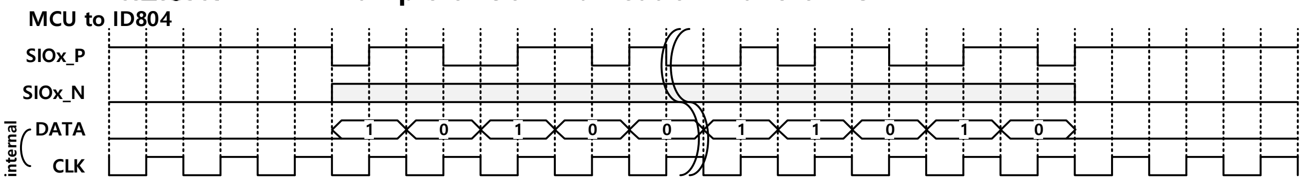


Figure 5. Example of MCU to ID804 (RX) communication

The receiving clock could be inverted via setup register (SETUP2.CLK_INV_E)

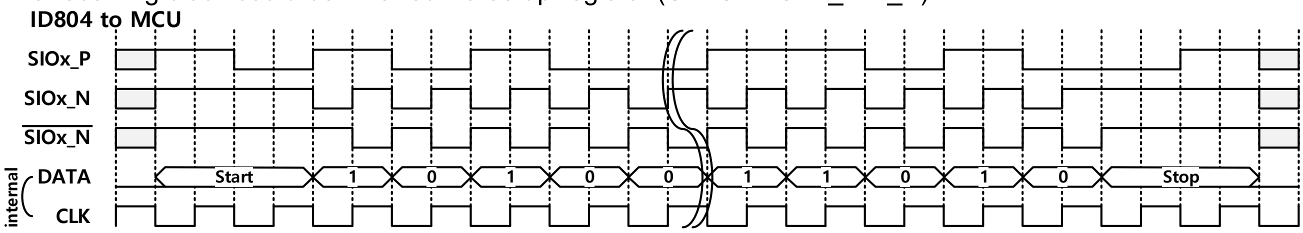


Figure 6. Example of ID804 to MCU (TX) communication

4.2.4. LVDS-mode communication physical layer and bit coding

For differential communication the termination resistance R_{LVDS_term} is 200 Ohm, differential voltage V_{LVDS_diff} is typ. 300 mV, data rate is typ. 2 Mbps. Common mode voltage V_{LVDS_comm} is typ.1.2 V, it means the max voltage on the communication lines during communication is typ. 1.35 V and the min. voltage is typ. 1.05 V

4.2.4.1. Example of Communication Waveforms

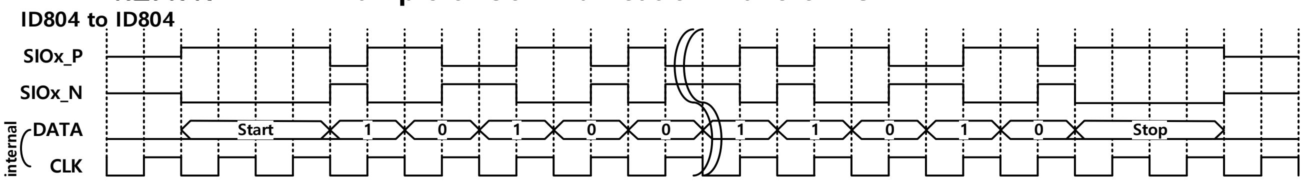


Figure 7. Example of ID804 to ID804 communication

4.2.5. CAN-mode communication physical layer and bit coding

If a transceiver is detected on SIOx the output and input message is via single ended & Manchester encoded communication with typ. 2 Mbps data transfer rate. For the transceiver the data transfer rate depends on the data signal due to the Manchester encoded data. Means up to the double transfer rate is needed (4 Mbps). Therefore, the use of a 5Mbps CAN-FD transceiver is required.

4.2.5.1. Example of Communication Waveforms

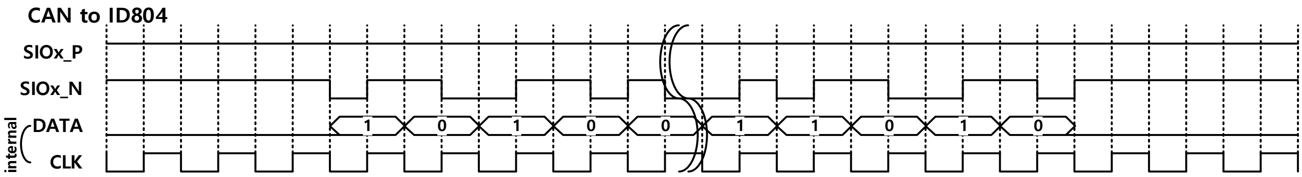


Figure 8. Example of CAN Transceiver to ID804 communication

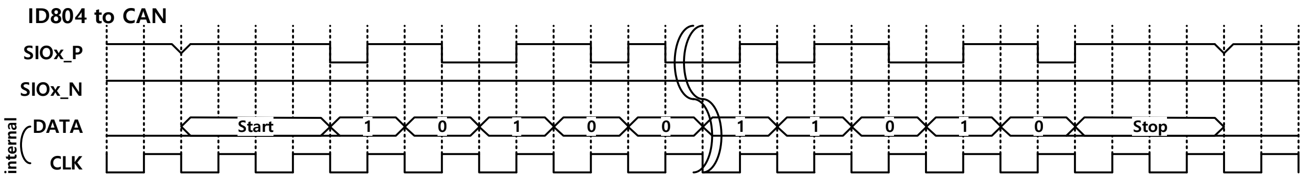


Figure 9. Example of ID804 to CAN Transceiver communication

4.2.6. Message Frame Format

ID804 has 3 types of frame format according to DATA bit. If the data bits are 12 or 24 bits, the upstream and downstream communication formats are the same. The 0-bit data frame format [Figure 12] can only be used for downstream communication.

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
55 54 53 52 51 50 49 48	47 46 45 44 43 42 41 40	39 38 37 36 35 34 33 32	31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
PREAMBLE (55–52), DEV_ADDR [11:0] (51–40)		COMMAND [7:0] (39–32)	DATA [23:0] (31–8)			CRC [7:0] (7–0)

Figure 10. 24bit Data Frame Format

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6
47 46 45 44 43 42 41 40	39 38 37 36 35 34 33 32	31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
PREAMBLE (47–44), DEV_ADDR [11:0] (43–32)		COMMAND [7:0] (31–24)	DATA [11:0] (23–12)	DUMMY [3:0] (11–8)	CRC [7:0] (7–0)

Figure 11. 12bit Data Frame Format

Byte 1	Byte 2	Byte 3	Byte 4
31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
PREAMBLE (31–28), DEV_ADDR [11:0] (27–16)		COMMAND [7:0] (15–8)	CRC [7:0] (7–0)

Figure 12. 0bit Data Frame Format

The bit order is MSB first.

- PREAMBLE (4 bit) : Fixed to 4'b1010.
  Avoid messages with incorrect preamble as this could lead to forwarding of garbage telegrams along the chain and to raising the STATUS2.COM_FLT.
- DEV_ADDR (12 bit) : Allowed values for downstream messages are 0x000 (broadcast) and 0x001 to 0xFEE (unicast nodes) and 0xFEF to 0xFFE (multicast). Upstream messages always contain the device address of the sender in the DEV_ADDR field. Not all commands are allowed in a broadcast message. See Chapter 4.5
- COMMAND (8 bit) **:** See Chapter 4.5
- DATA (24bit or 12bit or 0bit) : DATA field has bits according to COMMAND. See Chapter 4.5

- DUMMY (4bit) : In case of 12bit Data Frame Format, 4bit DUMMY data (4'b1010) inserted to match the data size to BYTE.
- CRC (8 bit) : See Chapter 4.2.7.8

4.2.7. Message Handling

4.2.7.1. Broadcast

An address of 0x000 initiates a broadcast, i.e., the same message is sent and received by all nodes on the chain. Whether a command can be broadcast is detailed in Chapter 4.5.

Broadcast Read or SR command is first received by all devices via the frame in downstream direction. The last node in the chain then immediately transmits its response frame upstream. The response frame's data field depends on the actual command. The response frame's address field is set according to the own device's address. All the nodes upstream forward all received response frames until a frame with the address of their adjacent node is received. Then the respective node transmits its own response frame. This procedure lasts until the chain's first node has transmitted its response frame.

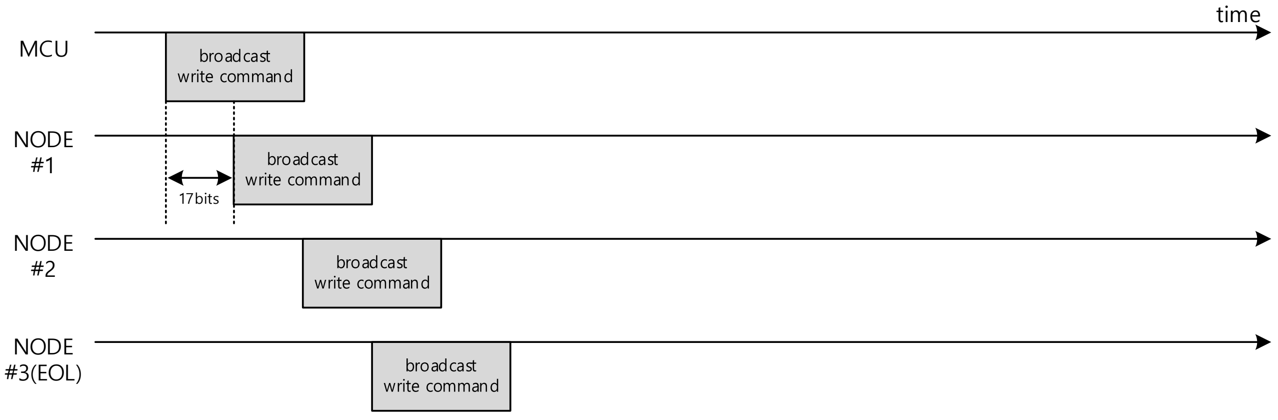


Figure 13. Example of broadcast write

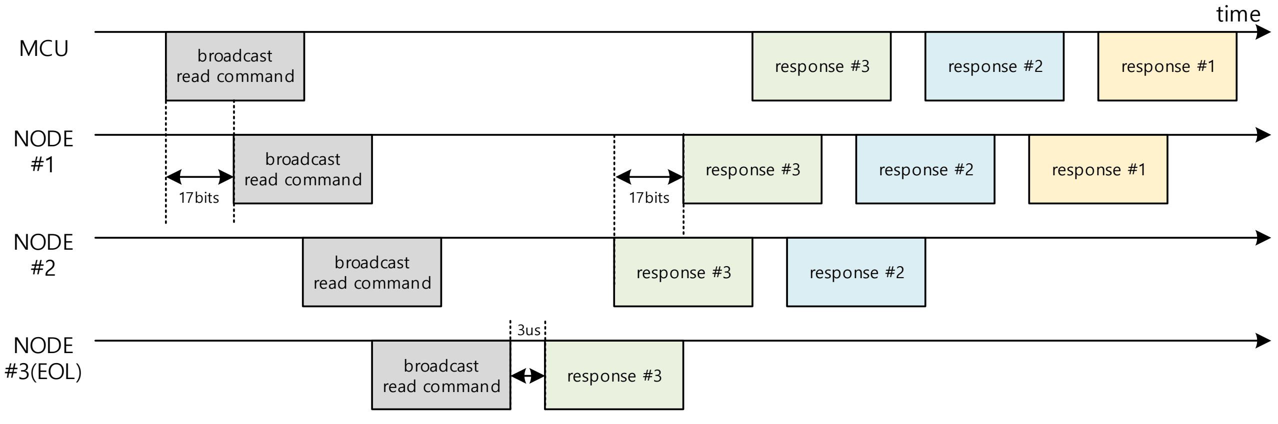


Figure 14. Example of broadcast read(or SR)

4.2.7.2. Multicast

An address of 0xFEF to 0xFFE initiates a multicast, i.e., the same message is sent and received by same defined group nodes on the chain. Multicast group can be defined by MCAST command. Whether a command can be multicast is detailed in Chapter 4.5.

4.2.7.3. Forwarding and Fast Forwarding

Message forwarding refers to the direct forwarding of a message without modification.
Messages are forwarded to the next device if the address in the message frame is the broadcast address, or multicast address, or it is different from the node's address.
If a message is to be forwarded, forwarding starts exactly one cycle after the preamble and address field are fully received, i.e., after 17 bits. This is referred to as fast-forwarding.
When a node is uninitialized, broadcast messages are fast-forwarded and multicast(0xFEF to 0xFFE) & unicast(0x001 to 0xFEE) & abnormal(0xFFF) addressed messages are forwarded only after the full telegram has been received.

4.2.7.4. Responses and SR (Status Requests)

Some commands trigger a response from the addressed node. In this case, the address field contains the address of the sending node to indicate the origin of the message. Once the node receives the end of the command frame, **the corresponding response will occur approximately 2us (in case of broadcast) or 26us (in case of unicast) later.**
Some commands do not send a response but allow a SR(Status Request). The SR(Status Request) is done by specific command. See Chapter 4.5. In case of broadcast response or SR see Chapter 4.2.7.1.

If a valid SR(Status Request) command has been received, the node responds by sending the TEMP & STATUS2 registers to the MCU after the original command has been executed. Status bits are cleared after responding to the SR(Status Request). In the case of the CLRERROR command, the status bits are cleared by the command before the status request is sent.
Commands supporting a status request are marked in the command description. See Chapter 4.5.

Note that the ID804 does not allow simultaneous upstream and downstream telegrams on the same node, as this could lead to dropped or corrupted messages.

4.2.7.5. Waiting Time between Telegrams

ID804 is not accepting a new telegram while a telegram is being received or transmitted on the other port. Likewise, simultaneous upstream and downstream messages are not supported.
Therefore, a waiting time must be added by the MCU between sending two telegrams in order to ensure that all receiving and transmitting activities have been concluded.
The minimum waiting time between two downstream telegrams should be 6 μs (edge to edge) and applies to write-only or without response request commands. If an answer has been requested, the next telegram should only be sent after the response has been received by the MCU.

4.2.7.6. Communication Errors

The ID804 requires certain communication errors to be detected by each device in a daisy chain. The below Table 2 summarizes all relevant error conditions.

Table 2. Communication Errors

IC State	Preamble	Address	Command	CRC	Non-Target DEV		Target DEV	
					COM_FLT flag	Data Forwarding	COM_FLT flag	Data Forwarding
Uninitialized state (UNINIT)	Incorrect	-	-	-	Set	Stop	Set	Stop
	Correct	Unicast	-	-	Set	No stop	-	-
		Multicast	-	-	Set	No stop	-	-
		Broadcast	Incorrect	-	-	-	Set	No stop
			Correct	Incorrect	-	-	Set	No stop
Initialized State (SLEEP, NORMAL, DEEPSLEEP)	Incorrect	-	-	-	Set	Stop	Set	Stop
	Correct	Unicast	Incorrect	-	No set	No stop	Set	Stop
			Correct	Incorrect	No set	No stop	Set	Stop
		Multicast	Incorrect	-	No set	No stop	Set	No stop
			Correct	Incorrect	No set	No stop	Set	No stop
		Broadcast	Incorrect	-	-	-	Set	No stop
			Correct	Incorrect	-	-	Set	No stop

The ID804 performs some error detection even for messages that are to be forwarded (CRC). If an error is detected in such a case, the COM_FLT flag is set and the message is still forwarded, i.e., communication is not stopped.

In UNINIT state(Uninitialized state), only broadcast address(0x000) and limit set of commands can accept "Correct" message. Other address (0x001 ~ 0xFFF) messages are ignored. And still forwarded and set COM_FLT flag.

Errors due to incorrect command is only detected if the device is addressed by unicast or multicast or broadcast.

4.2.7.7. **Communication Timeout**

The READ commands or SR WRITE commands trigger a response from the addressed or all nodes. Each node needs to await all responses originating from the nodes downstream. Thereafter either the node's own response is transmitted, or new commands are accepted. If new command(downstream) is received during response waiting, it is ignored and COM_FLT flag set on received node. Only the last node in the LED chain may immediately transmit its response. In case there is an error with the chain, not all expected responses may arrive. Thus, each of the commands expecting a response waits for a certain time only and then returns to its previous state without having transmitted the node's response data. The lengths of the timeouts depend on register setting command (SET_TIMEOUT). They are calculated to account for the worst-case oscillator frequency tolerance. I.e. the waiting node has a high-speed clock and all the nodes waited to have a low-speed clock. The hardware implementation uses an internal clock for the timeout counter. The resolution is internal clock (Fosc_main(Typ. 16MHz)) * (2^{10}).

4.2.7.8. **Cyclic Redundancy Check**

The CRC field contains the CRC8 check word calculated over the full message excluding the CRC field.
CRC Implementation: $x^8 + x^5 + x^3 + x^2 + x + 1$.
Polynomial: 0x2F (normal notation)
Initial value: 0x00
XOR value: 0x00

Direction: MSB first

Note that this implementation uses the same polynomial but is not compatible with the AUTOSAR implementation because of the different initial and XOR values of 0xFF.

Note: The calculation direction for the CRC field in the non-volatile memory is LSB first.

4.3. Color Control and Temperature Monitor

4.3.1. PWM Generation

The ID804 incorporates three independent PWM channels, one for each LED.
The PWM dimming resolution internally controlled by the IC is 12bit. The supported duty cycles are 0/4096 to 4095/4096. The PWM output frequency is set by SETUP1.F_PWM_DIV register setting. The frequency is decided by (Fosc_main(Typ. 16MHz)) / ((F_PWM_DIV + 1) X 4096). The PWM frequency is reduced to the quarter of set frequency when RGB data ≤ decimal 3 and the PWM frequency is reduced to the half of set frequency when decimal 4 ≤ RGB data ≤ decimal 7.
The output frequency is determined independently for each of the PWM channels.

4.3.1.1. PWM Update

When a new PWM duty cycle must be applied, this is always done at the end of a PWM duty cycle. I.e. the PWM always completes an output cycle using the previously active-duty cycle and starts the next output cycle using the updated duty cycle.

4.3.1.2. Phase Shift

To spread the current consumption of the LEDs over time, a phase shift enabled between the three PWM channels. It retains even if the output frequency of the channels is different.
The fixed phase shift is defined in the following Table 3. Please note the absolute phase shift times are nominal values. I.e. they are subject to vary with the internal oscillator's frequency.

Table 3. PWM Phase Shift

PWM Channel	Related Phase Shift	Unit
Green	0	degree
Red	90	degree
Blue	270	degree

4.3.2. Brightness Control Scaler

In operation, the color brightness of each LED is controlled via an 8-bit RGB register. This provides a color space of 3 x 8 bit = 24 bit. The 8-bit RGB value is scaled up to the 12-bit value using by Brightness calibration value(PWM_{MAX} value). The PWM_{MAX} value is stored in OTP.
The scaling equations of the red and green and blue channel are given in Table 4

Table 4. Scaling equation

Color	Scaling equation
Red	PWM_{RED} = (PWM_{MAX} X R_data + 128) >> 8
Green	PWM_{GREEN} = (PWM_{MAX} X G_data + 128) >> 8
Blue	PWM_{BLUE} = (PWM_{MAX} X B_data + 128) >> 8

4.3.3. Temperature Monitor

For temperature monitoring, the ID804 has a on-chip analog temperature sensor. This can be measured by ADC and value can be read from TEMPERATURE register through READ_TEMP command. This value allows monitoring the temperature of the device and compensating the brightness of the integrated LED based on temperature.

Table 5. Temperature equation

Parameter	Description	Equation	Unit
$T_{j,mon}$	junction temperature monitor	$\mathrm{Tj,mon} = 30℃ + \left(\frac{TEMPERATURE[9:0] - d440}{4.4}\right)$	℃
TEMPERATURE[9:0]	Junction Temperature value converted by ADC	10 bits	digit

4.4. Diagnostics

The ID804 provides built-in diagnostic functions for detection of fault conditions.
If a fault is detected, the respective fault flag bit is set. The fault flag bit can only be cleared when the STATUS2 register has been read by the MCU (either directly or via status request), all flag bits have been cleared using CLRERROR, or the device has received a RESET command.
If configured, the ID804 will automatically turn off the LED drivers when a failure is detected and change to SLEEP state to prevent damage to the part. This helps protect against device failures, ensuring communication is always possible throughout the daisy chain.

- Overtemperature (OT): This fault occurs when the IC junction temperature rises above the overtemperature threshold for more than 5 consecutive EVENT_CYC cycles time. It can only be cleared the temperature has fallen below the overtemperature release threshold.
- Undervoltage (UV): This fault occurs when the supply voltage(VDD) of the ID804 drops below the undervoltage threshold for more than 5 consecutive EVENT_CYC cycles time.
- LED open / short: This fault occurs when the LED driver pin voltage remains below the LED-open threshold voltage or above the LED-short threshold voltage for 16 consecutive FPWM_DIV cycles time. These fault conditions are tested individually for each channel, but only after the GOACTIVE command has been received and the PWM value of the channel is set to a value higher than the minimum value.
- Communication Fault : See Chapter 4.2.7.6.
- Communication Timeout : See Chapter 4.2.7.7

4.4.1. Fault Reaction

Table 6. Fault Reaction Table

Fault	Detection	Fault bit	Configuration	Reaction
VDD UV	VDD < V_{uvth_lo}	UV_FLT	UV_FLT_E=0	Only Fault bit set
			UV_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
OT	T_J > Temperature threshold	OT_FLT	OT_FLT_E=0	Only Fault bit set
			OT_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
LED OPEN	$V_{DRV\_R/G/B} < V_{TH\_OPEN\_lo}$ (Typ 160mV)	OPEN_R / OPEN_G / OPEN_B	OS_FLT_E=0	Only Fault bit set
			OS_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
LED SHORT	$V_{DRV\_R/G/B} > V_{TH\_SHORT}$ (Typ 4.4V)	SHORT_R / SHORT_G / SHORT_B	OS_FLT_E=0	Only Fault bit set
			OS_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
COMMUNICATION ERROR	See Table 2	COM_FLT	COM_FLT_E=0	Only Fault bit set
			COM_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
COMMUNICATION Timeout	See 4.2.7.7	T_OUT	-	Only Fault bit set

4.5. ID804 Commands

The ID804 supports the following commands.

Table 6. ID804 Commands

Command	Normal Command			SR (Status Request) Command			Data bits
	Encoding	Broadcast possible	Multicast possible	Encoding with SR	Broadcast possible	Multicast possible	
RESET	0xB1	O	O		-	-	0
INITBIDIR	0xB2	X	X		-	-	0
CLRERROR	0xB3	O	O	0xF3	O	X	0
GOSLEEP	0xB4	O	O	0xF4	O	X	0
GOACTIVE	0xB8	O	O	0xF8	O	X	0
GODEEPSLEEP	0xBC	O	O	0xFC	O	X	0
SET_SETUP1	0x86	O	O	0xC6	O	X	12
SET_SETUP2	0x87	O	O	0xC7	O	X	12
SET_MCAST	0x88	X	X	0xC8	X	X	24
SET_RGB	0xA0	O	O	0xE0	O	X	24
SET_TEMPTH	0x8A	O	O	0xCA	O	X	12
SET_TEMPHYS	0x8B	O	O	0xCB	O	X	12
SET_CURR_MAX_LVL	0x8F	O	O	0xCF	O	X	12
SET_TIMEOUT	0x9A	O	O	0xDA	O	X	12
READ_STATUS1	0x41	O	X	-	-	-	12
READ_STATUS2	0x42	O	X	-	-	-	12
READ_TEMP	0x43	O	X	-	-	-	12
READ_TEMPST	0x44	O	X	-	-	-	24
READ_SETUP1	0x46	O	X	-	-	-	12
READ_SETUP2	0x47	O	X	-	-	-	12
READ_MCAST	0x48	O	X	-	-	-	24
READ_RGB	0x60	O	X	-	-	-	24
READ_TEMPTH	0x4A	O	X	-	-	-	12
READ_TEMPHYS	0x4B	O	X	-	-	-	12
READ_PWM_RED_VAL	0x4C	O	X	-	-	-	12
READ_PWM_GREEN_VAL	0x4D	O	X	-	-	-	12
READ_PWM_BLUE_VAL	0x4E	O	X	-	-	-	12
READ_CURR_MAX_LVL	0x4F	O	X	-	-	-	12
READ_TIMEOUT	0x5A	O	X	-	-	-	12

If the ID804 receives a command that is not listed above, a communication error is raised.

Refer to the application note for detailed information on each command.

5. Package Description

5.1. Package outline Dimension

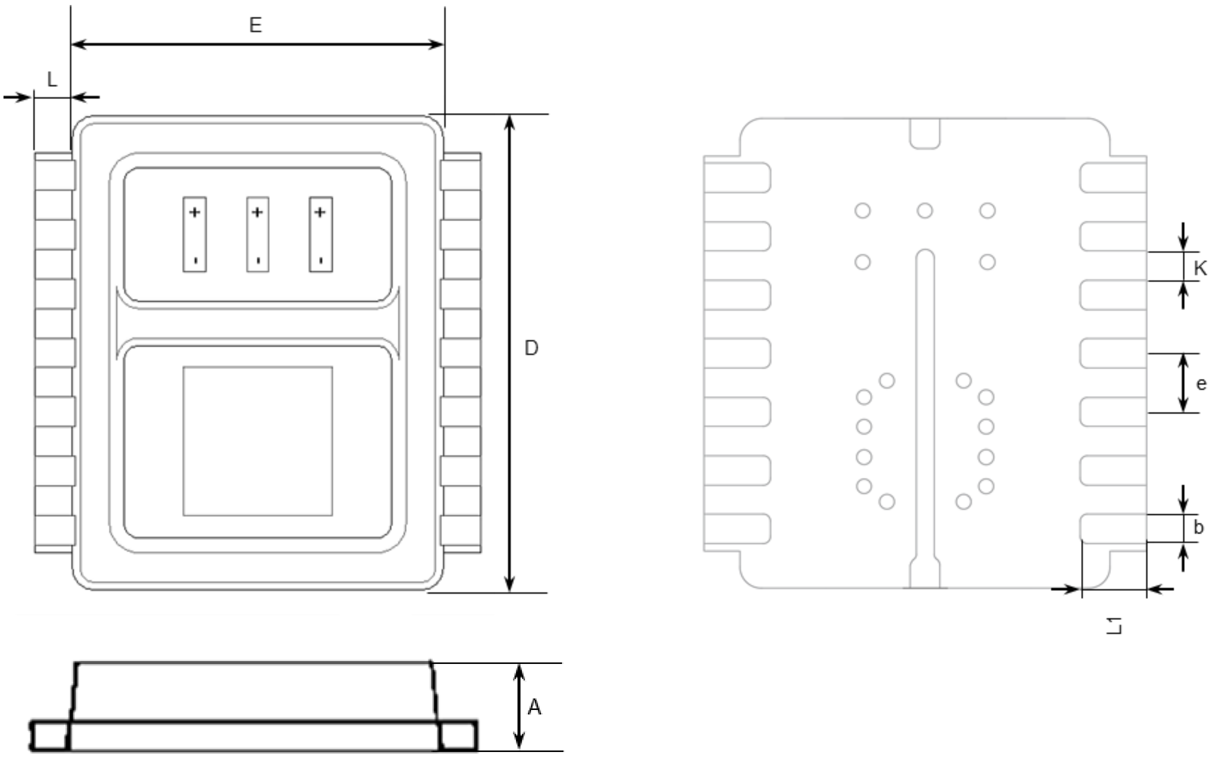
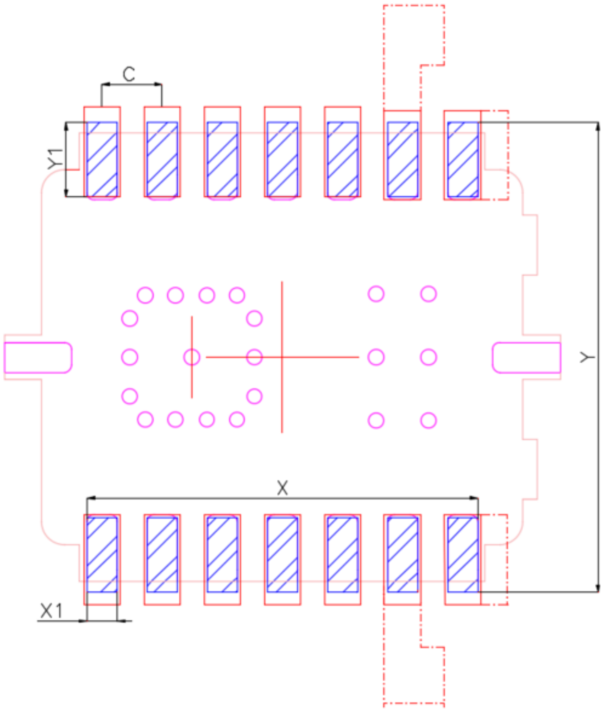


Figure 15. PoD (DFN 3225-14L)

Symbol	Dimensions in mm		
	Min	Nom	Max
A		0.65	
b	0.20 BSC		
D		3.21	
E		2.51	
e	0.40 BSC		
K	0.20 BSC		
L		0.245	
L1		0.45	

General tolerance is ±0.1mm.
RGB LED Pitch is 420um.

5.2. Recommended Solder Pad



Symbol	PKG	Solder Pad	Stencil
X	2.6	2.64	2.64
X1	0.2	0.24	0.24
Y	2.2	3.3	3.09
Y1	0.45	0.6	0.54
C	0.4	0.4	0.4

Note: The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD- 020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”